

Definition of Probability

Probability is a branch of mathematics that deals with the **likelihood of occurrence of an event**.

It quantifies uncertainty and provides a numerical measure between **0 and 1**, where:

- **0** represents an **impossible event**
- **1** represents a **certain event**

Classical Definition (Laplace's Definition)

If an experiment has **n** equally likely outcomes and an event **A** occurs in **m** of these outcomes, then the probability of event **A** is given by:

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}} = \frac{m}{n}$$

Example: Suppose you toss a fair coin. The total possible outcomes are **{Heads, Tails}** = 2. The probability of getting heads:

$$P(H) = \frac{1}{2}$$

Axiomatic Definition (Kolmogorov's Axioms)

According to Kolmogorov's probability theory, probability satisfies three axioms:

1. **Non-negativity:** $P(A) \geq 0$ for any event **A**
2. **Normalization:** $P(S) = 1$ (where **S** is the sample space)
3. **Additivity:** If **A** and **B** are mutually exclusive events, then $P(A \cup B) = P(A) + P(B)$

Empirical (Statistical) Definition

When an experiment is repeated **n** times and an event **A** occurs **m** times, the probability is:

$$P(A) = \lim_{n \rightarrow \infty} \frac{m}{n}$$

This approach is useful for real-world experiments where probabilities are estimated based on past data.

Example: If a cricket player has scored a century in **30 out of 100** matches, the probability of scoring a century in the next match is:

$$P(\text{Century}) = \frac{30}{100} = 0.3$$

Classical Approach of Probability (Laplace's Definition)

The **classical approach** of probability, also known as **Laplace's definition**, is based on the assumption that all possible outcomes of an experiment are **equally likely**. It is used when we have a **finite number of equally likely outcomes**.

Mathematical Definition

If an experiment has **n** equally likely outcomes and an event **A** occurs in **m** of these outcomes, then the probability of event **A** is:

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

Conditions for Applying Classical Probability

1. **Finite sample space:** The number of possible outcomes must be **countable** and **finite**.
2. **Equally likely outcomes:** Each outcome should have an **equal chance** of occurring.
3. **Mutually exclusive events:** The outcomes should not **overlap** (i.e., one outcome should not affect the occurrence of another).

Examples of Classical Probability

Example 1: Tossing a Fair Coin

When you toss a fair coin, there are **two possible outcomes**:

- **H (Heads)**
- **T (Tails)**

Since both outcomes are equally likely, the probability of getting a head is:

$$P(H) = \frac{1}{2}$$

Similarly, the probability of getting a tail is:

$$P(T) = \frac{1}{2}$$

Example 2: Rolling a Fair Die

A fair die has **6 faces** numbered {1, 2, 3, 4, 5, 6}. The total number of possible outcomes is **6**.

- Probability of rolling a **3**:

$$P(3) = \frac{1}{6}$$

- Probability of rolling an **even number (2, 4, or 6)**:

$$P(\text{even}) = \frac{3}{6} = \frac{1}{2}$$

Example 3: Drawing a Card from a Standard Deck

A standard deck has **52 cards**, with 13 cards in each suit (Hearts, Diamonds, Clubs, and Spades).

- Probability of drawing a **red card (Hearts or Diamonds)**:

$$P(\text{Red Card}) = \frac{26}{52} = \frac{1}{2}$$

- Probability of drawing an **Ace** (4 Aces in the deck):

$$P(\text{Ace}) = \frac{4}{52} = \frac{1}{13}$$

Limitations of Classical Probability

1. **Only works for finite sample spaces:** It cannot be used for experiments with **infinite** or **continuous** outcomes (e.g., measuring temperature or height).
2. **Assumption of equally likely outcomes is not always valid:** In real life, not all outcomes have the same probability (e.g., biased dice or loaded coins).
3. **Cannot handle complex probability distributions:** This approach does not apply to situations where outcomes follow different probability distributions (e.g., normal or Poisson distribution).

Empirical Approach of Probability (Statistical Probability)

The **empirical approach** of probability is based on **observations and experiments** rather than theoretical calculations. It is also called **statistical probability** or **experimental probability** because it relies on real-world data collected from repeated experiments.

Definition

If an experiment is repeated **n** times and an event **A** occurs **m** times, then the probability of event **A** is estimated as:

$$P(A) = \frac{\text{Number of times event A occurs}}{\text{Total number of trials}}$$

As the number of trials **n** increases, this probability **approaches the true probability** of the event.

Example 1: Tossing a Coin

If you toss a fair coin **100 times**, and you get **heads 48 times**, then the empirical probability of getting heads is:

$$P(H) = \frac{48}{100} = 0.48$$

This probability is **not exactly 0.5** because real-world experiments may have **small variations** due to randomness. However, if we repeat the experiment **many more times**, the probability will get closer to **0.5**.

Example 2: Rolling a Die

Suppose you roll a die **600 times**, and the outcomes are recorded as follows:

- 1 appears **92 times**
- 2 appears **105 times**
- 3 appears **98 times**
- 4 appears **102 times**
- 5 appears **110 times**
- 6 appears **93 times**

The empirical probability of rolling a **4** is:

$$P(4) = \frac{102}{600} = 0.17$$

If the die were perfectly fair, the probability of rolling any number would be **1/6 ≈ 0.1667**. However, real-world experiments often show small **deviations** due to imperfections.

Example 3: Weather Forecasting

If historical data shows that it rained on **70 out of 100** days when the humidity was above **80%**, then the empirical probability of rain on a high-humidity day is:

$$P(\text{Rain} \mid \text{Humidity} > 80\%) = \frac{70}{100} = 0.7$$

This probability is **based on past observations** and is used in weather forecasting.

Comparison: Classical vs. Empirical Probability

Feature	Classical Probability	Empirical Probability
Based on	Mathematical calculations (theory)	Experimental data (observations)
Requires equally likely outcomes?	Yes	No
Works for infinite trials?	No	Yes
Example	Tossing a fair die, drawing a card	Tossing a real die, weather predictions

Limitations of Empirical Probability

1. **Depends on sample size:** If the number of trials is small, the probability estimate may not be accurate.
2. **Can change over time:** Empirical probabilities are based on past data, which may not always represent future probabilities.
3. **Not useful for rare events:** If an event occurs very rarely (e.g., a **100-year flood**), we may not have enough data to estimate its probability correctly.

Axiomatic Approach of Probability (Kolmogorov's Axioms)

The **axiomatic approach** of probability was introduced by **Andrey Kolmogorov** in **1933**. Unlike the **classical** and **empirical** approaches, which rely on assumptions of equally likely outcomes or experimental data, the **axiomatic approach** defines probability in a rigorous mathematical way using **axioms** (fundamental rules).

This approach is **general and flexible**, allowing probability to be applied in various fields, including **machine learning, quantum physics, and economics**.

Kolmogorov's Axioms of Probability

Let **S** be the sample space of a random experiment, and let **A** be an event in **S**. The probability function $P(A)$ assigns a probability to every event and satisfies the following three axioms:

Axiom 1: Non-Negativity

For any event A , the probability is always **non-negative**:

$$P(A) \geq 0$$

Example: If rolling a fair die, the probability of getting any number (1, 2, 3, 4, 5, or 6) is always **greater than or equal to zero**.

Axiom 2: Total Probability of Sample Space

The probability of the entire sample space S is **always 1**:

$$P(S) = 1$$

Example: When rolling a fair six-sided die, the probability of getting **any one** of the six numbers (1, 2, 3, 4, 5, or 6) is:

$$P(1) + P(2) + P(3) + P(4) + P(5) + P(6) = 1$$

Axiom 3: Additivity (For Mutually Exclusive Events)

If A and B are **mutually exclusive events** (i.e., they cannot happen together), then:

$$P(A \cup B) = P(A) + P(B)$$

Example: If you roll a fair die, the probability of getting a **1** or a **2** is:

$$P(1) + P(2) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

Since rolling a **1** and rolling a **2** are mutually exclusive events, we **add** their probabilities.

Theorems Derived from Axioms

Using these axioms, we can derive many important probability properties:

Theorem 1: Probability of an Impossible Event is Zero

If A is an **impossible event** (i.e., it never happens), then:

$$P(A) = 0$$

Example: The probability of rolling a **7** on a fair six-sided die is **0**, because there is no **7** on the die.

Theorem 2: Complement Rule

For any event A , the probability that **A does not occur** (denoted as A^c , the complement of A) is:

$$P(A^c) = 1 - P(A)$$

Example: If the probability of raining today is **0.3**, then the probability of **not raining** is:

$$1 - 0.3 = 0.7$$

Theorem 3: Inclusion-Exclusion Rule

For **any two events** A and B (not necessarily mutually exclusive):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example: Suppose in a class of **100 students**:

- **40 students** study **Mathematics**.
- **50 students** study **Physics**.
- **20 students** study **both Mathematics and Physics**.

Then, the probability that a randomly chosen student studies **Mathematics or Physics** is:

$$\begin{aligned} P(\text{Math} \cup \text{Physics}) &= P(\text{Math}) + P(\text{Physics}) - P(\text{Math} \cap \text{Physics}) = \frac{40}{100} + \frac{50}{100} - \frac{20}{100} \\ &= \frac{70}{100} = 0.7 \end{aligned}$$

Comparison of Probability Approaches

Feature	Classical Approach	Empirical Approach	Axiomatic Approach
Based on	Equally likely outcomes	Observed experimental data	Fundamental mathematical rules
Formula	$P(A) = \frac{\text{Favorable outcomes}}{\text{Total outcomes}}$	$P(A) = \frac{\text{Observed frequency of A}}{\text{Total trials}}$	Based on Kolmogorov's axioms
Example	Rolling a fair die	Weather forecasting based on past data	Any probability model
Used in	Simple cases like cards, dice	Real-world events like medical trials	Advanced probability theory, ML, physics

Addition Theorem of Probability

The **addition theorem of probability** helps calculate the probability of **either** of two events occurring. It is particularly useful when dealing with **overlapping** or **mutually exclusive events**.

1. Addition Theorem for Two Events

For any two events A and B in the sample space S , the probability that **at least one of them occurs** is given by:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

where:

$P(A \cup B)$ = Probability of **A or B occurring** (either A, or B, or both).

- $P(A)$ = Probability of event A occurring.
- $P(B)$ = Probability of event B occurring.
- $P(A \cap B)$ = Probability of **both** A and B occurring (intersection).

Example 1: Tossing a Die

Consider rolling a fair six-sided die.

- Let event **A** = getting an even number $\{2,4,6\} \rightarrow P(A) = \frac{3}{6} = 0.5$.
- Let event **B** = getting a number greater than 3 $\{4,5,6\} \rightarrow P(B) = \frac{3}{6} = 0.5$.
- The intersection **A** $\cap$ **B** = getting both an even number **and** greater than 4,6 $\rightarrow P(A \cap B) = \frac{2}{6} = 0.33$.

Using the addition theorem:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.5 + 0.5 - 0.33 = 0.67$$

So, the probability of getting an **even number or a number greater than 3** is **0.67 (or 67%)**.

2. Addition Theorem for Mutually Exclusive Events

If **A and B are mutually exclusive** (they **cannot** happen together, i.e., $P(A \cap B) = 0$), the formula simplifies to:

$$P(A \cup B) = P(A) + P(B)$$

Example 2: Drawing a Card

A standard deck has **52 cards**.

- Let **A** be the event of drawing a **red card** (26 cards).
- Let **B** be the event of drawing a **black card** (26 cards).

Since **A and B cannot happen together** (a single card cannot be both red and black):

$$P(A \cup B) = P(A) + P(B) = \frac{26}{52} + \frac{26}{52} = 1$$

Thus, the probability of drawing **either a red or a black card** is **1 (100%)**, which makes sense because every card is either red or black.

3. Addition Theorem for Three Events

For three events A, B, C :

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

This formula adjusts for over-counting the intersections of pairs and the intersection of all three events.

Example 3: Survey on Favorite Drinks

A survey of **100 people** found:

- **40 people like Tea (T)** $\rightarrow P(T) = \frac{40}{100}$.
- **50 people like Coffee (C)** $\rightarrow P(C) = \frac{50}{100}$.
- **30 people like Juice (J)** $\rightarrow P(J) = \frac{30}{100}$.
- **10 people like both Tea and Coffee** $\rightarrow P(T \cap C) = \frac{10}{100}$.
- **5 people like both Coffee and Juice** $\rightarrow P(C \cap J) = \frac{5}{100}$.
- **8 people like both Tea and Juice** $\rightarrow P(T \cap J) = \frac{8}{100}$.
- **3 people like all three** $\rightarrow P(T \cap C \cap J) = \frac{3}{100}$.

Now, using the formula:

$$\begin{aligned} P(T \cup C \cup J) &= P(T) + P(C) + P(J) - P(T \cap C) - P(C \cap J) - P(T \cap J) + P(T \cap C \cap J) \\ &= 0.4 + 0.5 + 0.3 - 0.1 - 0.05 - 0.08 + 0.03 \\ &= 1 - 0.2 = 0.8 \end{aligned}$$

So, the probability of randomly selecting a person who likes at least one of these drinks is **0.8 (or 80%)**.

Key Takeaways

- ✓ **Addition theorem** is useful for finding the probability of at least one event occurring.
- ✓ If **events are mutually exclusive**, use $P(A \cup B) = P(A) + P(B)$.
- ✓ If **events overlap**, subtract $P(A \cap B)$ to avoid double-counting.
- ✓ For **three events**, adjust for **pairwise** and **triple** intersections.

Multiplication Theorem of Probability

The **multiplication theorem of probability** is used to determine the probability of the simultaneous occurrence of two or more events. It is based on the concept of **conditional probability**.

1. Multiplication Theorem for Two Events

For any two events A and B in a sample space S :

$$P(A \cap B) = P(A) \cdot P(B|A)$$

where:

- $P(A \cap B)$ = Probability of **both A and B occurring** (joint probability).
- $P(A)$ = Probability of event A occurring.
- $P(B | A)$ = Probability of event B occurring given that A has already occurred.

♦ Example 1: Drawing Two Cards from a Deck (Without Replacement)

Consider drawing **two cards** from a standard deck of **52 cards**.

- Let **A** be the event of drawing an **Ace first**. Since there are **4 Aces**, we have:

$$P(A) = \frac{4}{52} = \frac{1}{13}$$

- Let **B** be the event of drawing an **Ace second**, given that an Ace was drawn first. Now, only **3 Aces remain** in a deck of **51 cards**:

$$P(B|A) = \frac{3}{51}$$

Now, using the multiplication theorem:

$$P(A \cap B) = P(A) \cdot P(B|A) = \frac{1}{13} \times \frac{3}{51} = \frac{3}{663} = \frac{1}{221}$$

Thus, the probability of drawing **two Aces in a row** is **1/221 (or 0.00452, i.e., 0.452%)**.

2. Multiplication Theorem for Independent Events

Two events **A and B** are said to be **independent** if the occurrence of one does not affect the occurrence of the other.

For independent events, the formula simplifies to:

$$P(A \cap B) = P(A) \cdot P(B)$$

Example 2: Tossing Two Coins

Let event **A** be getting heads on the first coin, and event **B** be getting heads on the second coin.

Since tossing one coin **does not affect** the other, the events are independent.

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{2}$$

Using the multiplication theorem:

$$P(A \cap B) = P(A) \cdot P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

So, the probability of getting **two heads** is **1/4 (or 25%)**.

3. Multiplication Theorem for Three Events

For three events A, B , and C :

$$P(A \cap B \cap C) = P(A) \cdot P(B|A) \cdot P(C|A \cap B)$$

where:

- $P(A)$ = Probability of event A occurring.
- $P(B | A)$ = Probability of B occurring given that A has occurred.
- $P(C | A \cap B)$ = Probability of C occurring given that A and B have occurred.

Example 3: Drawing Three Cards (Without Replacement)

If we draw three cards from a deck without replacement,

- $P(A)$ = Probability of drawing a **King first** = $\frac{4}{52}$.
- $P(B | A)$ = Probability of drawing a **King second**, given a King was drawn first = $\frac{3}{51}$.
- $P(C | A \cap B)$ = Probability of drawing a **King third**, given two Kings were drawn = $\frac{2}{50}$.

$$\begin{aligned} P(A \cap B \cap C) &= P(A) \cdot P(B | A) \cdot P(C | A \cap B) \\ &= \frac{4}{52} \times \frac{3}{51} \times \frac{2}{50} = \frac{24}{132600} = \frac{1}{5525} \end{aligned}$$

Thus, the probability of drawing **three Kings in a row** is **1/5525 (or approximately 0.0181%)**.

Key Takeaways

✓ **Multiplication theorem** helps in finding the probability of multiple events happening together.

✓ For **dependent events**, use $P(A \cap B) = P(A) \cdot P(B | A)$.

✓ For **independent events**, use $P(A \cap B) = P(A) \cdot P(B)$.

✓ For **three or more events**, adjust the formula to account for conditional probabilities.

Bayes' Theorem of Inverse Probability

Bayes' Theorem is a fundamental concept in probability theory that describes how to update the probability of a hypothesis based on new evidence. It is also known as the **theorem of inverse probability** because it allows us to find the probability of a cause given an observed outcome.

1. Bayes' Theorem Formula

For two events **A** and **B**, Bayes' Theorem states:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

where:

- $P(A | B)$ = **Posterior Probability** (Probability of event A given that B has occurred).
- $P(B | A)$ = **Likelihood** (Probability of event B given that A has occurred).
- $P(A)$ = **Prior Probability** (Initial probability of event A before considering B).
- $P(B)$ = **Total Probability of B** (Sum of all possible ways B can occur).

If event B can occur due to multiple mutually exclusive events $A_1, A_2, \dots, A_n$, then:

$$P(B) = P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + \dots + P(B|A_n)P(A_n)$$

2. Example 1: Medical Diagnosis (False Positives and False Negatives)

Problem Statement:

A certain disease affects **1% of a population**. A diagnostic test for the disease is **90% accurate for positive cases** and **95% accurate for negative cases**. If a person tests positive, what is the probability that they actually have the disease?

Given Data:

- Let D = Person **has the disease**
- Let D^c = Person **does not have the disease**
- Let T^+ = **Positive test result**

We are given:

- $P(D) = 0.01$ (**Prior probability of having the disease**)

- $P(D^c) = 1 - P(D) = 0.99$ (**Prior probability of not having the disease**)
- $P(T^+|D) = 0.90$ (**Test detects disease correctly 90% of the time**)
- $P(T^+|D^c) = 0.05$ (**False positive rate: 5% of healthy people test positive**)

Using the **total probability rule**, we find $P(T^+)$:

$$\begin{aligned} P(T^+) &= P(T^+|D)P(D) + P(T^+|D^c)P(D^c) \\ &= (0.90 \times 0.01) + (0.05 \times 0.99) \\ &= 0.009 + 0.0495 \\ &= 0.0585 \end{aligned}$$

Now, applying **Bayes' Theorem**:

$$\begin{aligned} P(D|T^+) &= \frac{P(T^+|D)P(D)}{P(T^+)} P(D | T^+) \\ &= \frac{(0.90 \times 0.01)}{0.0585} \\ &= \frac{0.009}{0.0585} \\ &= 0.154 \end{aligned}$$

Conclusion:

Even though the test is 90% accurate, the probability that a person **actually has the disease given a positive test result is only 15.4%**. This is because the disease is rare, so false positives impact the final probability significantly.

3. Example 2: Spam Email Detection

A spam filter classifies emails as spam or not based on certain words. Suppose:

- 30% of emails are **spam** ($P(S) = 0.3$).
- 70% are **not spam** ($P(S^c) = 0.7$).
- The word "**free**" appears in **40% of spam emails** ($P(F | S) = 0.4$).
- The word "**free**" appears in **5% of non-spam emails** ($P(F|S^c) = 0.05$).

We want to find:

$$P(S|F) \quad (\text{Probability that an email is spam given it contains } free)$$

Using the total probability formula:

$$\begin{aligned}
 P(F) &= P(F|S)P(S) + P(F|S^c)P(S^c)P(F) \\
 &= (0.4 \times 0.3) + (0.05 \times 0.7) \\
 &= 0.12 + 0.035 = 0.155
 \end{aligned}$$

Applying **Bayes' Theorem**:

$$P(S|F) = \frac{P(F|S)P(S)}{P(F)} = \frac{0.4 \times 0.3}{0.155} = \frac{0.12}{0.155} = 0.774$$

Conclusion:

If an email contains the word "free", there is a **77.4% chance that it is spam**.

4. Key Insights from Bayes' Theorem

- ✓ Helps in **updating probabilities** when new evidence is available.
- ✓ Used in **medical testing, spam filtering, AI and machine learning, fault diagnosis**, and more.
- ✓ Demonstrates why **false positives matter** in medical tests and classification problems.
- ✓ Helps in **decision-making** where prior probabilities are updated based on observed data.

Properties of Probability with Proofs

Probability is a mathematical measure of uncertainty. If S is a **sample space** and A is an **event**, then the probability of A is denoted by $P(A)$. The properties of probability are derived from **Kolmogorov's Axioms**.

Kolmogorov's Axioms of Probability

A probability measure P satisfies the following three axioms:

1. Non-Negativity:

$$P(A) \geq 0 \quad \text{for any event } A$$

(Probability of any event is non-negative)

2. Normalization (Probability of Sample Space):

$$P(S) = 1$$

(The probability of the sample space is always 1)

3. Additivity (For Disjoint Events):

If A and B are **mutually exclusive** events, then:

$$P(A \cup B) = P(A) + P(B)$$

(The probability of the union of mutually exclusive events is the sum of their probabilities)

Property 1: Probability of the Complement of an Event

Statement:

The probability of the complement of an event A , denoted as A^c , is:

$$P(A^c) = 1 - P(A)$$

Proof:

From Kolmogorov's Axioms, we know that:

1. A and A^c are **mutually exclusive**.
2. $A \cup A^c = S$ (The event and its complement cover the entire sample space).

Applying the additivity axiom:

$$P(A \cup A^c) = P(A) + P(A^c)$$

Since $P(S) = 1$, we substitute:

$$1 = P(A) + P(A^c)$$

Rearranging,

$$P(A^c) = 1 - P(A)$$

Example:

If the probability of it **raining tomorrow** is 0.3, then the probability of it **not raining** is:

$$P(\text{Not Raining}) = 1 - 0.3 = 0.7$$

Property 2: Probability of the Union of Two Events (Addition Theorem)

Statement:

For any two events A and B :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof:

By the definition of probability, the union of two events consists of all elements in A or B , but the intersection $A \cap B$ is counted twice. To correct this:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example:

Suppose:

- Probability of selecting a **math book** = 0.4
- Probability of selecting a **science book** = 0.5
- Probability of selecting **both** = 0.2

Then,

$$P(\text{Math or Science}) = 0.4 + 0.5 - 0.2 = 0.7$$

Property 3: Probability of the Empty Set**Statement:**

The probability of the **empty event** (impossible event) is **zero**:

$$P(\emptyset) = 0$$

Proof:

Since the empty set contains no outcomes, we have:

$$S = A \cup A^c$$

Applying the probability axiom:

$$P(S) = P(A) + P(A^c)$$

Since $P(S) = 1$ and $A = \emptyset$, we get:

$$P(\emptyset) + P(A^c) = 1$$

Since $A^c = S$, we get:

$$P(\emptyset) = 0$$

Example:

The probability of selecting a **negative number** from the set $\{1,2,3,4,5\}$ is 0.

Property 4: Probability of a Subset (Monotonicity)**Statement:**

If $A \subseteq B$, then:

$$P(A) \leq P(B)$$

Proof:

Since $A \subseteq B$, we can write:

$$B = A \cup (B - A)$$

Since A and $B - A$ are mutually exclusive:

$$P(B) = P(A) + P(B - A)$$

Since $P(B - A) \geq 0$, we get:

$$P(A) \leq P(B)$$

Example:

- Probability of choosing a **red card** from a deck of 52 cards = $\frac{26}{52} = 0.5$
 - Probability of choosing **any card (red or black)** = 1
- Since red cards are a subset of all cards, $P(\text{Red}) \leq P(\text{Any Card})$.

Property 5: Multiplication Rule of Probability

Statement:

For any two events A and B , the probability of both occurring is:

$$P(A \cap B) = P(A)P(B|A)$$

If A and B are independent:

$$P(A \cap B) = P(A)P(B)$$

Proof:

From the definition of conditional probability:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Multiplying both sides by $P(A)$:

$$P(A \cap B) = P(A)P(B|A)$$

Example:

Suppose the probability of **choosing a blue ball** from a bag is 0.4, and given a blue ball was chosen, the probability of then **choosing a red ball** is 0.6.

$$P(\text{Blue} \cap \text{Red}) = P(\text{Blue})P(\text{Red}|\text{Blue}) = 0.4 \times 0.6 = 0.24$$

Summary Table of Probability Properties

Property	Formula
Complement Rule	$P(A^c) = 1 - P(A)$
Union of Two Events	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Property	Formula
Empty Set	$P(\emptyset) = 0$
Subset Rule	$P(A) \leq P(B) \text{ if } A \subseteq B$
Multiplication Rule	$(P(A \cap B) = P(A) P(B)$

Probability and Statistics by Prof. Aland S.R.